In [24]:
```
# mplement K-Means clustering/ hierarchical clustering on sales_data_sample.csv
# Determine the number of clusters using the elbow method. Dataset link :
```

In [25]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

In [26]:
```
df = pd.read_csv('sales_data_sample.csv', encoding = 'latin1')
```

In [27]:
```
print(df.info())
print(df.isnull().sum())
print(df.columns)
print(df.head())
```

```
<class 'pandas.DataFrame'>
RangeIndex: 2823 entries, 0 to 2822
Data columns (total 25 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   ORDERNUMBER       2823 non-null   int64
 1   QUANTITYORDERED   2823 non-null   int64
 2   PRICEEACH         2823 non-null   float64
 3   ORDERLINENUMBER   2823 non-null   int64
 4   SALES             2823 non-null   float64
 5   ORDERDATE         2823 non-null   str
 6   STATUS            2823 non-null   str
 7   QTR_ID            2823 non-null   int64
 8   MONTH_ID          2823 non-null   int64
 9   YEAR_ID           2823 non-null   int64
 10  PRODUCTLINE       2823 non-null   str
 11  MSRP              2823 non-null   int64
 12  PRODUCTCODE       2823 non-null   str
 13  CUSTOMERNAME      2823 non-null   str
 14  PHONE             2823 non-null   str
 15  ADDRESSLINE1      2823 non-null   str
 16  ADDRESSLINE2      302 non-null    str
 17  CITY              2823 non-null   str
 18  STATE             1337 non-null   str
 19  POSTALCODE        2747 non-null   str
 20  COUNTRY           2823 non-null   str
 21  TERRITORY         1749 non-null   str
 22  CONTACTLASTNAME   2823 non-null   str
 23  CONTACTFIRSTNAME  2823 non-null   str
 24  DEALSIZE          2823 non-null   str
dtypes: float64(2), int64(7), str(16)
memory usage: 551.5 KB
None
ORDERNUMBER            0
QUANTITYORDERED        0
PRICEEACH              0
ORDERLINENUMBER        0
SALES                  0
ORDERDATE              0
STATUS                 0
QTR_ID                 0
MONTH_ID               0
YEAR_ID                0
PRODUCTLINE            0
MSRP                   0
PRODUCTCODE            0
CUSTOMERNAME           0
PHONE                  0
ADDRESSLINE1           0
ADDRESSLINE2        2521
CITY                   0
STATE               1486
POSTALCODE            76
COUNTRY                0
TERRITORY           1074
CONTACTLASTNAME        0
CONTACTFIRSTNAME       0
DEALSIZE               0
dtype: int64
Index(['ORDERNUMBER', 'QUANTITYORDERED', 'PRICEEACH', 'ORDERLINENUMBER',
```

```
       'SALES', 'ORDERDATE', 'STATUS', 'QTR_ID', 'MONTH_ID', 'YEAR_ID',
       'PRODUCTLINE', 'MSRP', 'PRODUCTCODE', 'CUSTOMERNAME', 'PHONE',
       'ADDRESSLINE1', 'ADDRESSLINE2', 'CITY', 'STATE', 'POSTALCODE',
       'COUNTRY', 'TERRITORY', 'CONTACTLASTNAME', 'CONTACTFIRSTNAME',
       'DEALSIZE'],
      dtype='str')
```

	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER	SALES	\
0	10107	30	95.70	2	2871.00	
1	10121	34	81.35	5	2765.90	
2	10134	41	94.74	2	3884.34	
3	10145	45	83.26	6	3746.70	
4	10159	49	100.00	14	5205.27	

	ORDERDATE	STATUS	QTR_ID	MONTH_ID	YEAR_ID	...	\
0	2/24/2003 0:00	Shipped	1	2	2003	...	
1	5/7/2003 0:00	Shipped	2	5	2003	...	
2	7/1/2003 0:00	Shipped	3	7	2003	...	
3	8/25/2003 0:00	Shipped	3	8	2003	...	
4	10/10/2003 0:00	Shipped	4	10	2003	...	

	ADDRESSLINE1	ADDRESSLINE2	CITY	STATE	\
0	897 Long Airport Avenue	NaN	NYC	NY	
1	59 rue de l'Abbaye	NaN	Reims	NaN	
2	27 rue du Colonel Pierre Avia	NaN	Paris	NaN	
3	78934 Hillside Dr.	NaN	Pasadena	CA	
4	7734 Strong St.	NaN	San Francisco	CA	

	POSTALCODE	COUNTRY	TERRITORY	CONTACTLASTNAME	CONTACTFIRSTNAME	DEALSIZE
0	10022	USA	NaN	Yu	Kwai	Small
1	51100	France	EMEA	Henriot	Paul	Small
2	75508	France	EMEA	Da Cunha	Daniel	Medium
3	90003	USA	NaN	Young	Julie	Medium
4	NaN	USA	NaN	Brown	Julie	Medium

[5 rows x 25 columns]

In [28]:
```
df
```

Out[28]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>ORDERNUMBER</th> <th>QUANTITYORDERED</th> <th>PRICEEACH</th> <th>ORDERLINENUMBER</th> <th>SALES</th> <th>ORDERDATE</th> <th>STATUS</th> <th>QTR_ID</th> <th>MONTH_ID</th> <th>YEAR_ID</th> <th>...</th> <th>ADDRESSLINE1</th> <th>ADDRESSLINE2</th> <th>CITY</th> <th>STATE</th> <th>POSTALCODE</th> <th>COUNTRY</th> <th>TERRITORY</th> <th>CONTACTLASTNAME</th> <th>CONTACTFIRSTNAME</th> <th>DEALSIZE</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>...</th> <th>2818</th> <th>2819</th> <th>2820</th> <th>2821</th> <th>2822</th> </tbody> <tbody></tbody>

10107	30	95.70	2	2871.00	2/24/2003 0:00	Shipped	1	2	2003	...	897 Long Airport Avenue		NYC	NY	10022	USA		Yu	Kwai	Small
10121	34	81.35	5	2765.90	5/7/2003 0:00	Shipped	2	5	2003	...	59 rue de l'Abbaye		Reims		51100	France	EMEA	Henriot	Paul	Small
10134	41	94.74	2	3884.34	7/1/2003 0:00	Shipped	3	7	2003	...	27 rue du Colonel Pierre Avia		Paris		75508	France	EMEA	Da Cunha	Daniel	Medium
10145	45	83.26	6	3746.70	8/25/2003 0:00	Shipped	3	8	2003	...	78934 Hillside Dr.		Pasadena	CA	90003	USA		Young	Julie	Medium
10159	49	100.00	14	5205.27	10/10/2003 0:00	Shipped	4	10	2003	...	7734 Strong St.		San Francisco	CA		USA		Brown	Julie	Medium
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
10350	20	100.00	15	2244.40	12/2/2004 0:00	Shipped	4	12	2004	...	C/ Moralzarzal, 86		Madrid		28034	Spain	EMEA	Freyre	Diego	Medium
10373	29	100.00	1	3978.51	1/31/2005 0:00	Shipped	1	1	2005	...	Torikatu 38		Oulu		90110	Finland	EMEA	Koskitalo	Pirkko	Medium
10386	43	100.00	4	5417.57	3/1/2005 0:00	Resolved	1	3	2005	...	C/ Moralzarzal, 86		Madrid		28034	Spain	EMEA	Freyre	Diego	Medium
10397	34	62.24	1	2116.16	3/28/2005 0:00	Shipped	1	3	2005	...	1 rue Alsace-Lorraine		Toulouse		31000	France	EMEA	Roulet	Annette	Small
10414	47	65.52	9	3079.44	5/6/2005 0:00	On Hold	2	5	2005	...	8616 Spinnaker Dr.		Boston	MA	51003	USA		Yoshido	Juri	Medium

2823 rows × 25 columns

In [29]:
```
# QUANTITYORDERED        SALES                PRICEEACH
x = df[['QUANTITYORDERED','SALES','PRICEEACH']]
from sklearn.preprocessing import StandardScaler
```

```
sc = StandardScaler()
x = sc.fit_transform(x)
```

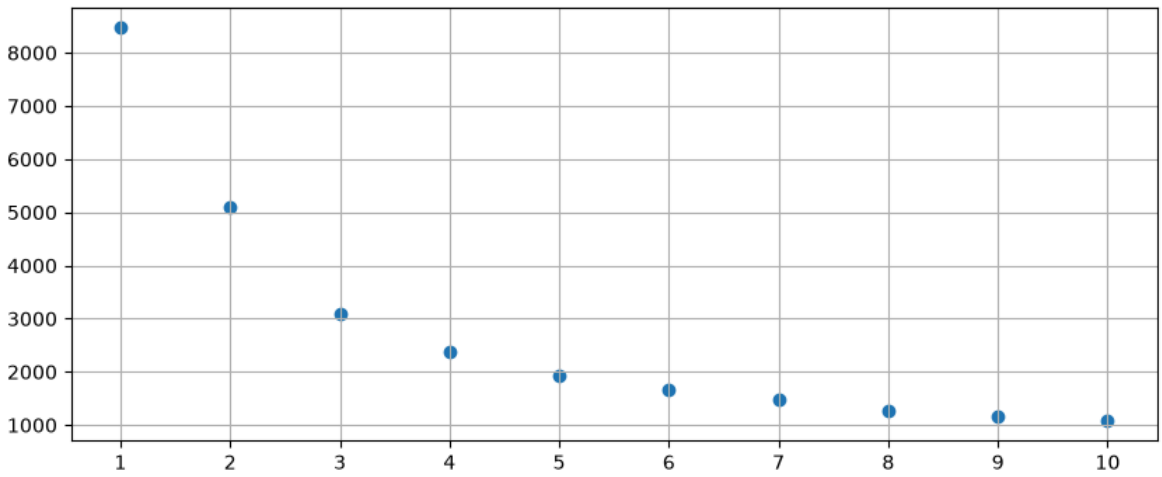
In [30]:
```
from sklearn.model_selection import train_test_split
from sklearn.cluster import KMeans
```

In [31]:
```
possible_k = []

for i in range(1,11):
    km = KMeans(n_clusters=i)
    km.fit(x)
    possible_k.append(km.inertia_)
```

In [32]:
```
plt.figure(figsize=(10,4))
plt.scatter(range(1,11), possible_k, alpha=True)
plt.grid(True)
plt.xticks(range(1,11))
```

Out[32]:
```
([<matplotlib.axis.XTick at 0x21443a6da90>,
  <matplotlib.axis.XTick at 0x21443b00050>,
  <matplotlib.axis.XTick at 0x21443b007d0>,
  <matplotlib.axis.XTick at 0x21443b00f50>,
  <matplotlib.axis.XTick at 0x21443b016d0>,
  <matplotlib.axis.XTick at 0x21443b01e50>,
  <matplotlib.axis.XTick at 0x21443b025d0>,
  <matplotlib.axis.XTick at 0x21443b02d50>,
  <matplotlib.axis.XTick at 0x21443b034d0>,
  <matplotlib.axis.XTick at 0x21443b03c50>],
 [Text(1, 0, '1'),
  Text(2, 0, '2'),
  Text(3, 0, '3'),
  Text(4, 0, '4'),
  Text(5, 0, '5'),
  Text(6, 0, '6'),
  Text(7, 0, '7'),
  Text(8, 0, '8'),
  Text(9, 0, '9'),
  Text(10, 0, '10')])
```



In [33]:
```
k = 3
km = KMeans(n_clusters=k, random_state=42)
pred = km.fit_predict(x)
```

In [34]:
```
pred
```

Out[34]:
```
array([2, 2, 1, ..., 1, 0, 0], shape=(2823,), dtype=int32)
```

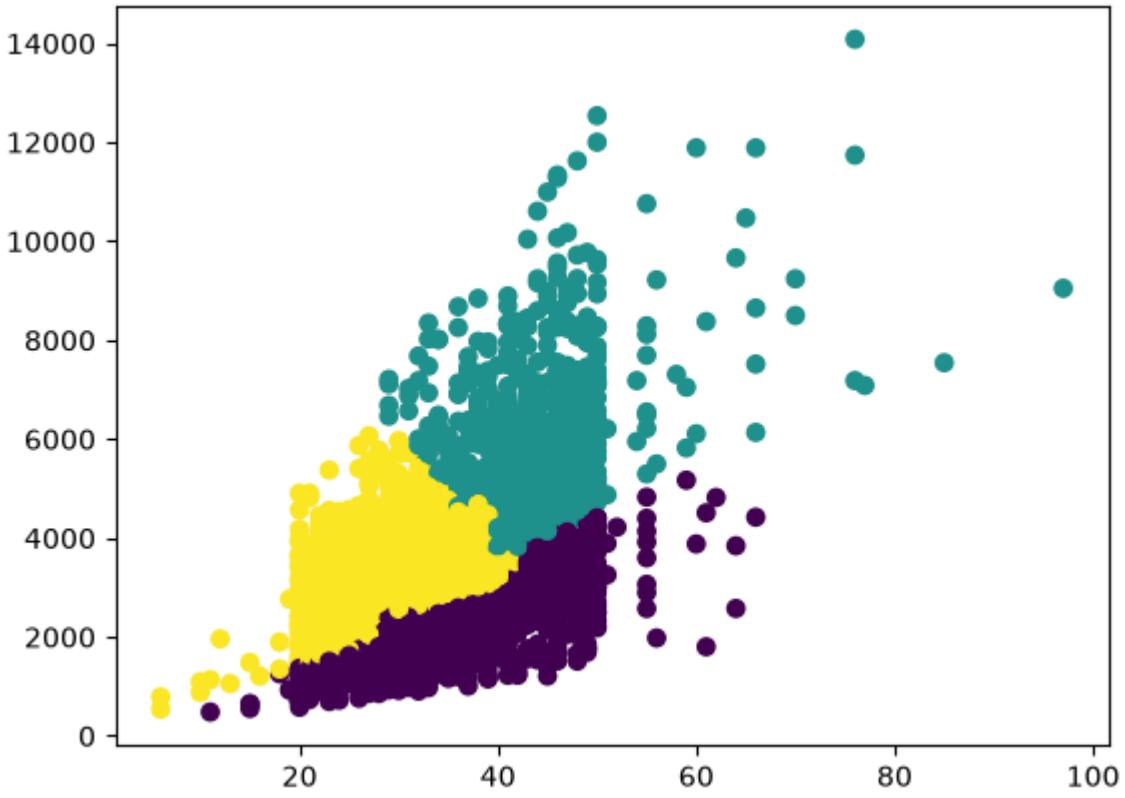
In [35]:
```
from sklearn.cluster import AgglomerativeClustering
ag = AgglomerativeClustering(
    n_clusters=k,
    linkage='ward'
)

clust = ag.fit_predict(x)

df['cluster'] = clust
```

In [40]:
```
plt.scatter(df["QUANTITYORDERED"],df["SALES"], c=df['cluster'])
```

Out[40]:
```
<matplotlib.collections.PathCollection at 0x21443ce1f90>
```



In [ ]:
```

```